Effects of temperature on reproduction in *Stentor coeruleus*

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Introduction

Temperature is known to affect various biological processes across lifeforms. Previous studies have shown that temperature can affect reproduction mechanisms in several ciliate species.

Previously, we observed effects of temperature on *Stentor coeruleus* movement and regeneration.

This led us to investigate temperature influences on stentor division. We decided to test stentors at 10, 20, and 30 degrees celsius.

Material & Methods

- *Stentor coeruleus* from a local lake kept at room temperature and fed twice a week with algae.
- We set three incubators to 10°C, 20°C and 30°C
- We prepared 9 petri dishes with 25 mL spring water, 15 stentors each, and fed 50 µL of algae solution. With 3 dishes in each incubator.
- 3x/week (Tue, Thur, Sat) for 3 weeks, we counted # stentors per dish and fed 50 µL algae.
- Using an ANOVA test, we compared the means of each of the three temperature groups.

Results

The 20°C and 30°C groups increase rapidly in the first ten days (Fig 1, green, red). The 10°C group increased slightly (Fig 1, blue).

ANOVA test: statistically significant difference between 10 and 20 and 10 and 30 (Fig 1, $p = 0.008643$).

The 30°C group averaged 2.3833 cell divisions, 20°C 2.2833, and 10°C 0.7778 (Fig 2).

Figures

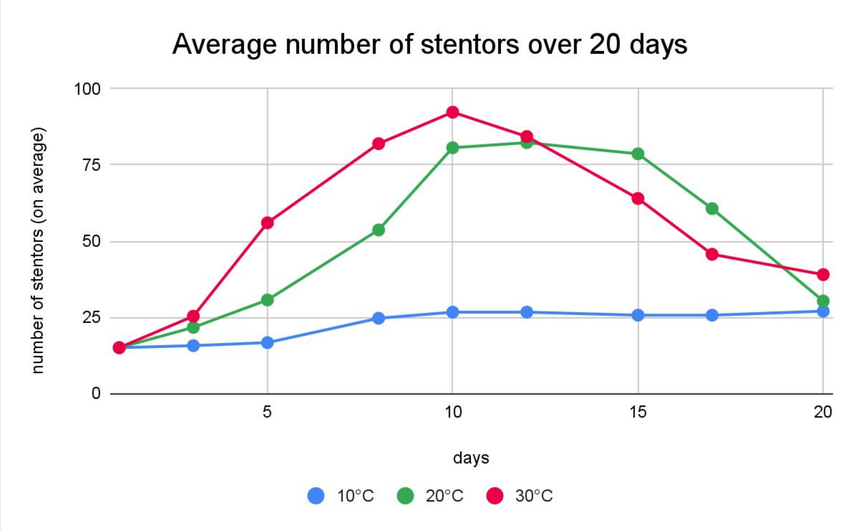

Figure 1: Number of Stentors Present according to Temperature
The blue line corresponds to the data collected for the 10°C group, the green is 20°C, and the red is 30°C. Each line represents the average of three groups of data per temperature. The data is shown over a period of twenty days with nine data points. The ANOVA test outputted a p value of 0.008643, suggesting a statistically significant difference between the 10°C group and both the 20°C and 30°C, but no statistically significant difference was found between the 20°C and 30°C groups.

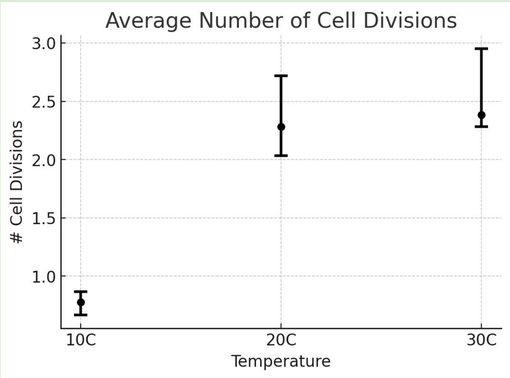

Figure 2: Average number of cell divisions over 10 days
The black dots represent the average cell divisions per temperature. The top error bar represents the most amount of cell divisions, and the bottom error bar indicates the smallest number of cell divisions. Average cell divisions represented by dots: 10°C = 0.7778, 20°C = 2.2833, 30°C = 2.3833.

Figure 3: Image of melted stentor
This is an image taken through a microscope of a melted stentor after being exposed to 30°C for an extended period of time.

Figure 4: Image of stentor in 30°C
This is an image of a stentor taken through a microscope. We observed dark spots that may indicate the consumption of smaller cells by larger stentors.

Discussion

This experiment looks at the effects of temperature on stentor reproduction by comparing the rates of cell division at 10°C, 20°C, and 30°C. According to our previous experiments, we expected to see a decrease in division rates under 10°C conditions.

Stentors kept at ~25°C are expected to divide 1-2x/week. For the first 10 days, the stentors divided at reasonable rates: 2.3833x for 30°C, 2.2833x for 20°C, and 0.778x for 10°C, (Fig 2).

After 10 days, stentors in 20°C and 30°C declined. Possible reasons for this include heat exposure (30°C) and lack of resources (food, space, oxygen) (Fig 3,4). In order to better understand what caused the dramatic decrease in cells, we could try different feeding dosages and closely monitoring what stentors consume.

Researching potential effects of temperature on stentor division is helpful for further understanding how temperature might play a role in the cellular reproduction of various other organisms. This is relevant due to the ever more concerning matter of climate change. Drastic shifts in temperature across the planet may have more profound effects than anticipated.

Bibliography

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